Convergent Recognition Theory: A Geometric Coherence Functional

for Information Geometry

J. Sellers (Oakwood, OH)

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Conventions

All logarithms are base-2. All entropies are in **bits**. We write log for $\log_2$ throughout. The von Neumann entropy is S(rho) = -Tr(rho log rho) in bits. The Shannon entropy is H(P) = -SUM P_i log P_i in bits. The symbol M denotes the set of model frameworks $\{M_1,...,M_n\}$; the quantum statistical manifold is denoted Q.

Abstract

*We present a **geometric coherence functional** Psi in [0, 1) defined on the product space (Q x Q) x (model space) x (apparatus space), where Q is the quantum statistical manifold. Psi is the product of a **Disjoint Recognition Core** $R_{12} := F_{gated}$ (gated quantum fidelity between independent batch reconstructions) and a **Global Reliability Modulator** $G := C_{eff}$ * D-hat (apparatus-level consensus and signal quality). The multiplicative structure is constructed from an exponentiated effective action: Psi = 2^(-S_eff) = R_{12} * G, where R_{12} in [0, 1] and G in [0, 1). The Recognition Core $R_{12} = F_{gated}$ is a smooth, symmetric two-point function on the product manifold Q x Q. For a fixed model set and fixed redundancy partition, Psi is continuous on the resulting domain; globally, Psi is piecewise continuous due to two sources of jump discontinuity: (a) transitions in the active model set (eta threshold) and (b) transitions in the redundancy partition (n_{unique}). The regulated normalization JSD/(H(w) + H_{min}) ensures smooth continuation and introduces no additional jumps. Psi is experimentally implementable using standard quantum optics hardware with non-trivial tomographic computational pipelines. The **Recognition Term** (gated quantum fidelity) uses a smooth geometric-mean informativeness gate that penalizes shared ignorance. The **Convergence Coefficient** (normalized Jensen-Shannon divergence with redundancy control) is defined over measurement-outcome distributions from a fixed informationally complete POVM. The **Detection Quality Weight** uses a bounded sigmoidal map D-hat = D/(D + D_{ref}). We present two falsifiable predictions, two operational validations, and a concrete quantum optics experimental protocol. Finally, we present a speculative hypothesis in Appendix A linking the spacetime gradient of Psi to tidal curvature.*

1. Introduction

Similar cross-check structures appear in many inference settings: when two independent systems reconstruct a shared referent, their combined output contains information inaccessible to either alone. Here we formalize one operational instance in quantum optics by defining a **pipeline-consistency functional** Psi for reconstruction convergence.

We make two claims. The **weak claim** (well-defined and implementable): Psi is a rigorous, **operational pipeline-consistency functional** for a specified experiment and reconstruction stack. It is a well-defined, bounded, measurable functional on the product space (Q x Q) x (model space) x (apparatus space). R_{12} acts as the disjoint-batch recognition core, measuring strict independent agreement. G serves as a global, pooled-data apparatus reliability prior. The weak claim has immediate testable predictions in quantum optics. The **strong claim** (speculative hypothesis, see Appendix A): the spacetime variation of Psi correlates with spacetime curvature.

Every definition has been tightened to eliminate dimensional ambiguity, non-differentiability, and unbounded behavior. Version history is provided in Appendix B.

2. The Exponentiated Action (Principled Ansatz)

Psi is constructed as a **principled log-additive ansatz**. The functional decomposes into a **Disjoint Recognition Core** $R_{12} := F_{gated}$ (gated fidelity between independent batch reconstructions) and a **Global Reliability Modulator** $G := C_{eff}$ * D-hat (model consensus times signal quality). The multiplicative structure follows from exponentiating the sum of independent failure-mode penalties. Define the **effective action**:

$$\mathbf{S_{eff} := -\log F_{gated} - \log C_{eff} - \log \text{D-hat}}$$

(Eq. 1)

2.1 Axiomatic Design Requirements

The functional form is selected to satisfy the following operational axioms: (1) **Boundedness:** Psi in [0, 1). (2) **Symmetry:** $F_{gated}(rho_1, rho_2) = F_{gated}(rho_2, rho_1)$. (3) **Monotonicity:** Psi increases monotonically in fidelity, model consensus, and SNR. (4) **Zero-locus:** For the ideal theoretical functional Psi, the zero-locus is strict when F = 0 (orthogonal states) or D-hat = 0 (no signal). In the implemented estimator Psi-hat, full-rank regularization, the epistemic gate, and the H_{min} numerical guard impose strictly positive, albeit negligible, numerical floors characterized by O(eps, $delta_{reg}$, H_{min}). High-entropy agreement is penalized to the eps floor (~eps/(1+eps)), but not to zero. Model disagreement (C_{eff}) approaches but never reaches zero under regulated normalization. The log-additive exponentiated action is a **natural, mathematically simple choice** satisfying these operational axioms. **Remark:** Assuming continuity and multiplicativity of independent penalties, Cauchy's functional equation motivates the exponential map. We select this log-additive form over alternatives (e.g., generalized means, t-norms, or min/max hybrids) for its analytical separability and its ability to cleanly isolate the distinct failure modes of fidelity, consensus, and signal strength.

Domain and Continuous Extension: S_{eff} is defined on the open interior of the product space where all three factors are strictly positive ($F_{gated} > 0$, $C_{eff} > 0$, D-hat > 0). On the **zero set** of the multiplicative factors (e.g., orthogonal states where F = 0, or no signal where D-hat = 0), S_{eff} -> +infinity and Psi is **continuously extended** to 0: lim Psi = lim 2^(-S_eff) = 0 as S_{eff} -> infinity. Note that zeros can occur in the interior of state space (e.g., orthogonal states), not only at the boundary. For a fixed model set and fixed redundancy partition, Psi is continuous on the resulting domain. Globally, Psi is **piecewise continuous** due to two sources of jump discontinuity: (a) transitions in the active model set n_{active} (models crossing the weight threshold eta), and (b) transitions in the redundancy partition n_{unique} (clustering threshold crossings). The regulated normalization JSD/(H(w) + H_{min}) is smooth and introduces no additional jumps. Within each region of parameter space where the discrete partition (the active set A and n_{unique}) remains constant, Psi is smooth and continuous. In finite-sample implementations, the estimated functional Psi-hat is piecewise continuous globally due strictly to the discrete selection rules of the active-set thresholding and redundancy graph partitioning. While the regulated normalization in Eq. 9 is smooth with respect to its inputs (JSD_w, H(w)), these discrete rules introduce jump discontinuities. In the population limit with fixed thresholds, Psi is strictly continuous on each region. F_{gated} is C-infinity on the full-rank interior, strictly excluding the set where F = 0. At F = 0, the action S_{eff} diverges, and Psi is defined strictly by continuous extension to 0.

where each operand is dimensionless and bounded: $R_{12} = F_{gated}$ in [0, 1], C_{eff} in (0, 1], D-hat in [0, 1). Since -log is non-negative on (0, 1], S_{eff} >= 0 whenever all factors are positive. The **coherence functional** is:

$$\mathbf{Psi := 2^{(-S_eff)} = R_{12} * G = F_{gated} * C_{eff} * \text{D-hat}}$$

(Eq. 2 - Central Result)

The product form follows from the additive structure of the action via the exponential map (base 2, matching our bit convention). This mirrors the common technique of converting additive information-theoretic penalties into multiplicative weights via exponentiation.

Global bound: F_{gated} in [0, 1], C_{eff} in (0, 1], D-hat in [0, 1). Therefore Psi lies in **[0, 1)**. Psi = 0 when any factor vanishes. Psi approaches 1 only in the simultaneous limit of perfect fidelity, perfect convergence, and infinite signal-to-noise. In all physical experiments, Psi < 1.

3. Formal Definitions

3.1 Recognition Term: Gated Quantum Fidelity F_{gated}

Let rho_1, rho_2 be density operators in a d-dimensional Hilbert space, reconstructed via state tomography. Define:

$$\mathbf{F(rho_1, rho_2) := (Tr\ sqrt(sqrt(rho_1)\ rho_2\ sqrt(rho_1)))^2\ [Uhlmann\ fidelity]}$$

(Eq. 3)

$$\mathbf{I(rho) := 1 - S(rho) / log(d)\ [Informativeness,\ S\ in\ bits]}$$

(Eq. 4)

Here d is the **fixed Hilbert space dimension** of the reconstruction model, determined a priori by the physical measurement setup (e.g., d = 4 for a two-qubit bipartite system, $d = d_A * d_B$ for general bipartite states). This is not a free parameter.

$$\mathbf{G_{eps}(rho_1, rho_2) := sqrt((I(rho_1) + eps)(I(rho_2) + eps)) / (1 + eps)\ [Normalized\ smooth\ gate]}$$

(Eq. 5)

$$\mathbf{F_{gated} := F(rho_1, rho_2) * G_{eps}(rho_1, rho_2)}$$

(Eq. 6)

where eps > 0 is a fixed regularization constant (eps = 10^{-8} in numerical implementations). The division by (1 + eps) ensures G_{eps} <= 1 exactly, preventing F_{gated} from exceeding 1. The geometric mean replaces the min() gate from v5, ensuring G_{eps} is C-infinity on the interior of state space. This permits the gradient computations required by the GR bridge hypothesis.

Bounds: F in [0, 1] (standard). I(rho) in [0, 1] since 0 <= S(rho) <= log(d). The numerator of G_{eps} ranges from eps (both I = 0) to 1 + eps (both I = 1). After dividing by (1 + eps), G_{eps} lies in [eps/(1+eps), 1] which is a subset of (0, 1]. Therefore F_{gated} lies in **[0, 1]** exactly. The gate ensures that the informativeness weight G_{eps} remains bounded in [eps/(1+eps), 1], preventing shared ignorance from causing numerical instability while allowing F_{gated} to correctly track vanishing fidelity. The informativeness gate intentionally enforces a positive **epistemic floor** (~eps). The use of von Neumann entropy to gate shared ignorance is a **deliberate epistemic weighting design choice**, not an operationally derived theorem. It intentionally penalizes high-entropy agreement to enforce a strict definition of informative recognition. Maximal agreement between maximally mixed states represents a lack of usable information, and is thus treated as operationally uninformative rather than an ideal convergence. In the theoretical limit where As purity approaches maximum (I -> 1), the gate G_{eps} -> 1 and F_{gated} -> F. The numeric floor eps/(1+eps) vanishes strictly in the limit as eps -> 0.

Full-rank regularization: Reconstruction algorithms (particularly MLE) may produce rank-deficient density matrices with zero eigenvalues. While entropy is formally defined on the boundary via convention, numerical stability and gradient differentiability degrade at rank-deficient points. To ensure all estimators remain strictly on the interior of the state space, all reconstructed density matrices undergo a full-rank regularization step before entering the functional: rho-tilde = (1 - $delta_{reg}$) rho + $delta_{reg}$ I/d, where $delta_{reg}$ = 10^{-6} and I/d is the maximally mixed state. This guarantees that all eigenvalues are strictly positive, ensuring the functional remains in the stable C-infinity interior. The perturbation on the informativeness gate is strictly bounded by a continuous function f($delta_{reg}$, d) governed by the Audenaert-Fannes inequality [19], ensuring the bias is negligible. While the remaining factors (F, C_{eff}, D-hat) are individually continuous under trace distance, we do not assert a global Lipschitz bound for the full product functional here. The numerical sensitivity table (Appendix B.1) confirms that the total perturbation remains negligible (|Delta Psi| < 10^{-5}) for $delta_{reg}$ = 10^{-6} across all test states.

3.2 Convergence Coefficient: C_{eff}

Let M = {M_1,...,M_n} be n **model frameworks**, i.e., the set of tomographic reconstruction estimators (e.g., MLE, Bayesian MAP, compressed sensing, linear inversion). Each model M_i is applied to the same tomographic dataset using a fixed **informationally complete POVM**, producing a predicted measurement-outcome distribution P_i over the POVM outcome space. JSD is defined on these classical distributions. Let w_{raw} = ($w_{raw,1}$,...,$w_{raw,n}$) be the initial normalized model weights (summing to 1 over all n models). The raw weights are defined via a softmax mapping over the cross-validated log-likelihoods: $w_{raw,i}$ = exp(LL_i / tau) / SUM_j exp(LL_j / tau), where the temperature parameter tau = 1 is fixed by convention. Define the **active set** A = { i : $w_{raw,i}$ >= eta }, where eta = 10^{-6} is the active weight threshold (default), and n_{active} = |A|. Define **renormalized active weights** w_i := $w_{raw,i}$ / ($sum_{j\ in\ A}$ $w_{raw,j}$), ensuring the active weights form a valid probability vector (sum to 1). All subsequent information-theoretic quantities (P_{mix}, JSD_w, H(w)) are computed using w_{active} = $(w_i)_{i\ in\ A}$. Transitions in the active set (models crossing the eta threshold) contribute to the piecewise continuity of Psi alongside n_{unique} transitions. **Invariance note:** Only JSD_{norm} is invariant under the addition of zero-weight models. The full coefficient C_{eff} is intentionally not invariant, as the ratio R = n_{unique}/n_{active} actively penalizes redundancy and depends on eta-thresholding.

Define:

$$\mathbf{P_{mix} := SUM_i\ w_i\ P_i}$$

(Eq. 7)

$$\mathbf{JSD_w := H(P_{mix}) - SUM_i\ w_i\ H(P_i)\ [in\ bits]}$$

(Eq. 8)

Notation: We use $P_i(k)$ = Tr(E_k rho_i) as the predicted outcome distribution; in Algorithm 1, this is denoted $P_{rec,i}$ for clarity. P_i and $P_{rec,i}$ refer to the same mathematical object.

The upper bound of JSD_w for n distributions with weights w is H(w) (the Shannon entropy of the weight vector). The bound JSD_w <= H(w) follows from the concavity of Shannon entropy and Jensen's inequality applied to the weighted mixture (see Lin 1991 [3] for the exact weighted multi-distribution proof). To ensure numerical stability as H(w) -> 0, we utilize the **regulated normalization**:

$$\mathbf{JSD_{norm} := JSD_w / (H(w) + H_{min})\ [in\ [0,\ 1)]}$$

(Eq. 9)

where $H_{min} = 10^{-6}$ is the numerical floor. When H(w) = 0 (single nonzero weight), $JSD_w = 0$ and $JSD_{norm} = 0$. For n = 1, we define $JSD_{norm} = 0$ and $C_0 = 1$ by convention (single-framework limit). This regulated normalization by $H(w) + H_{min}$ rather than $\log_2(n)$ ensures invariance under the addition of irrelevant (zero-weight) models, prevents division-by-zero artifacts, and introduces no additional jump discontinuity. While the regulator is smooth in its arguments (JSD_w, H(w)), the mapping from raw weights to these quantities remains piecewise due to active-set thresholding. Define raw convergence:

$$\mathbf{C_0 := 1 - JSD_{norm} \ [in \ [0, 1]]}$$

(Eq. 10)

To prevent redundant models from inflating convergence, we introduce an explicit redundancy equivalence class via a bootstrap-calibrated procedure:

Algorithm 1: Bootstrap-Calibrated Redundancy Clustering

1. Fix B = 1000 bootstrap iterations. Each predicted outcome distribution is defined as $P_{rec,i}(k) = Tr(E_k \ rho_i)$ under the fixed IC-POVM $\{E_k\}$. For each iteration b: resample the coincidence data with replacement (**resampling unit:** outcome counts per POVM setting block — block bootstrap); run two independent tomographic reconstructions (utilizing varied starting seeds) to estimate estimator-internal variability under finite sampling; compute $JSD(P_{rec,1}, P_{rec,2})$. For convex solvers (e.g., projected-gradient MLE), this internal variability serves as a diagnostic check and should be negligible.

2. Set $delta_i$ = 95th percentile of the B pairwise JSD values for estimator family i. The thresholds $delta_i$ are finite-sample estimators of the true population 95th percentile under the measurement model, derived via the B = 1000 bootstrap. This is the statistical floor below which two outputs are indistinguishable from finite-sample noise. The tolerance $delta_i$ captures the combined spread from sampling variability and estimator stochasticity for family i. Model weights w_{raw} are computed once from the full original dataset (via the softmax mapping over cross-validated log-likelihoods); bootstrap iterations are used strictly to estimate the family-specific thresholds $delta_i$. Cross-validation splits for weight computation are fixed by a published random seed to ensure deterministic reproducibility of the model weights w_{raw}. Both the cross-validation splits and the B = 1000 bootstrap resampling utilize a fixed, published RNG seed to ensure the deterministic reproducibility of the $delta_i$ thresholds and n_{unique}.

3. Construct a **redundancy graph** G over the active models (i in A). An edge exists between M_i and M_j if $JSD(P_i, P_j) < delta_{ij}$, where $delta_{ij} = max(delta_i, delta_j)$. The redundancy classes are unambiguously defined as the **connected components** of this graph, intentionally allowing transitive closure to define broad equivalence classes. n_{unique} := number of connected components. This definition is uniquely determined, deterministic, and computable in $O(n_{active}^2)$ time. **Computational Scope:** The $O(n_{active}^2)$ distance matrix computation is polynomial. The bootstrap (B = 1000) introduces a large constant factor. We explicitly target small model ensembles ($n_{active} < 10$) typical in lab tomography, where this runtime is trivial.

4. Define the redundancy ratio:

$$\mathbf{R := n_{unique} / n_{active} \ [in \ [1/n_{active}, 1]]}$$

(Eq. 11)

The effective convergence is:

$$\mathbf{C_{eff} := C_0 * R \ [in \ (0, 1]]}$$

(Eq. 12)

C_{eff} is intentionally not invariant under model set augmentation; it incorporates an explicit **redundancy penalty** (duplicate suppression factor) via n_{unique}/n_{active} to discourage redundant model ensembles. Adding duplicate or near-duplicate estimators decreases C_{eff} whenever the duplicate falls within the bootstrap-calibrated equivalence tolerance (see Lemma 2).

Lemma 1 (Bounds on C_{eff})

Since $JSD_w <= H(w)$ (Lin 1991 [3]), it follows that $JSD_{norm} = JSD_w/(H(w) + H_{min})$ in [0, 1) for all w, hence $C_0 = 1 - JSD_{norm}$ in (0, 1]. Since $1 <= n_{unique} <= n_{active}$, the ratio $R = n_{unique}/n_{active}$ lies in $[1/n_{active}, 1]$. Therefore $C_{eff} = C_0 * R$ lies in (0, 1]. Because of the regulated normalization, C_{eff} does not reach 0. At maximal disagreement ($JSD_w -> H(w)$), it approaches the positive numerical floor: $C_{eff} -> (H_{min} / (H(w) + H_{min})) * R$. QED.

Because of the regulated normalization, exact zero convergence is mathematically prohibited. Conceptually, even maximal disagreement among bounded models yields a strictly positive, albeit negligible, epistemic weight. This is a feature, not a bug: the functional distinguishes 'total disagreement' from 'no data' (the latter is captured by D-hat = 0).

Lemma 2 (Redundancy Monotonicity in the Ideal Duplicate Limit)

Assuming the duplicate model's output lies within delta of its original (as expected under the bootstrap calibration), n_{unique} remains constant while n_{active} increases by 1. Since $C_{eff} = C_0 * R$, the ratio $n_{unique}/(n_{active}+1) < n_{unique}/n_{active}$, and C_{eff} strictly decreases. QED.

3.3 Detection Quality Weight: Bounded Sigmoidal Map D-hat(e)

Let D(e) be the experimentally measured **operational signal density** (e.g., coincidence counts per time window) and D_{ref} be the empirical reference density (detector noise floor, e.g., accidental coincidence rate). Define:

$$\textbf{D-hat(e) := D(e) / (D(e) + } \mathbf{D_{ref}} \textbf{)}$$

(Eq. 13)

Properties: D-hat is bounded in **[0, 1)**. D-hat = 0 when D(e) = 0 (no signal). D-hat -> 1 as D(e) -> infinity (saturating). D-hat = 0.5 when $D(e) = D_{ref}$ (signal equals noise). D-hat is C-infinity for $D(e) >= 0$ and $D_{ref} > 0$, monotonically increasing in D(e), and dimensionless (ratio of densities). The action term -log(D-hat) is defined only for D > 0, with a continuous extension to Psi = 0 as D -> 0.

Geometric interpretation: $-\log \text{D-hat} = -\log(D/(D+D_{ref})) = \log(1 + D_{ref}/D)$, which is the penalty incurred by the noise floor relative to signal. This maps to the third term of the effective action.

Operational surrogate: D-hat is an operational signal-to-noise weight. Under a fixed measurement alphabet and time window, the count rate serves as an operational surrogate for detection quality (see Section 9.2). This is an operational penalty term, not derived from an optimal coding theorem unless a specific detection-likelihood model is provided. Psi is a flux-sensitive functional: D-hat varies with photon detection rate and directly modulates the global output.

3.4 Parameter Sensitivity

The parameters eps (informativeness gate) and $delta_{reg}$ (full-rank regularization) are numerical stabilizers whose sole function is to enforce the C-infinity interior of the state space. Sensitivity analysis indicates that for small values ($eps = 10^{-8}$, $delta_{reg} = 10^{-6}$), variations in these parameters have a negligible effect on the functional's output for non-degenerate states.

Lemma 3 (Global Bounds on Psi)

*F_{gated} in [0, 1], C_{eff} in (0, 1], D-hat in [0, 1). Therefore Psi = F_{gated} * C_{eff} * D-hat lies in **[0, 1)**. Psi = 0 when any factor vanishes. Psi approaches 1 only in the simultaneous limit of perfect fidelity, perfect convergence, and infinite signal-to-noise. In all physical experiments, Psi < 1. QED.*

4. Theoretical Justifications

The log-additive ansatz treats each component as an independent penalty isolating a distinct failure mode. Each component is chosen for specific mathematical and operational reasons:

Uhlmann Fidelity (F): Chosen for its direct relationship to the Bures metric on the quantum statistical manifold and its operational interpretation as the maximum overlap between purifications, giving it a natural role in quantifying statistical indistinguishability between reconstructed states.

Shannon Informativeness (I): Chosen because the von Neumann entropy is a canonically characterized additive measure of information uncertainty on quantum states under standard axioms (Shannon-Khinchin axioms generalized to the quantum case; see Wehrl 1978, up to a multiplicative constant). The informativeness I = 1 - S/log(d) linearly maps maximal entropy to zero informativeness.

Jensen-Shannon Divergence (JSD): For n = 2 distributions, sqrt(JSD) is a metric (Endres & Schindelin 2003); we utilize the weighted multi-distribution generalization for model ensembles. JSD provides a bounded, symmetric divergence that generalizes naturally to weighted model ensembles.

Detection Quality Weight (D-hat): Chosen as the simplest dimensionless saturating map for signal-to-noise weighting. The form $D/(D + D_{ref})$ is a canonical dimensionless ratio monotone, bounded in [0, 1), and equals 0.5 at the noise floor.

5. Consistency Checklist

Property	F_gated	C_eff	D-hat	Psi
Bounds	[0, 1]	[0, 1]	[0, 1)	[0, 1)
Dimensionless	Yes	Yes	Yes	Yes
Smooth (C-inf)	Yes (eps-reg, normalized)	Depends on model class & weight rule	Yes (D_ref>0)	Yes on open interior
Log base	n/a	base-2 (bits) JSD/(H(w)+H_min)	n/a	base-2 action
Defined on	Density ops	Classical P_i	Real D>=0	Product
Measurable	Tomography	POVM + models	Count rates	Product

6. Information Geometry

The quantum statistical manifold Q of density operators, equipped with the Bures metric, is a Riemannian manifold with computable curvature (Braunstein & Caves 1994, Petz 1996). Formally, Psi lives on a **product space**: (Q x Q) x (model space) x (apparatus space). The Recognition Term F_{gated} is a smooth, symmetric **two-point function** on Q x Q; it becomes a scalar field only after the selection of a fixed reference state rho_*, yielding $F_{gated}(rho, rho_*)$. The Convergence Coefficient C_{eff} depends on the discrete model set M and weight rule, and the Detection Quality Weight D-hat depends on the apparatus parameters. The effective action S_{eff} thus defines a function on the full product space. A speculative hypothesis connecting the spatial variation of F_{gated} to spacetime curvature is presented separately in Appendix A.

We define a **reconstruction divergence penalty** (a monotone transform of gated fidelity used for penalty weighting, not a formal metric):

$$\mathbf{d_{rec}^{\,2} := -\log F_{gated}}$$

(Eq. 14)

This is non-negative (since F_{gated} in (0, 1]) and vanishes when $F_{gated} = 1$. It is monotonically related to fidelity but is explicitly not the Bures distance and does not satisfy the triangle inequality.

6.1 Structural Analogies

Quantum Mechanics: n = 1 ($C_{eff} = 1$), Psi reduces to a noise-gated, signal-weighted fidelity measure. Operationally equivalent to the signal quality metric in quantum error correction.

Bayesian Inference: Classical limit: F -> Bhattacharyya coefficient, C_{eff} measures redundancy-controlled model agreement, D-hat maps to a bounded evidence-to-noise ratio. Psi parallels the posterior in Bayesian model averaging.

Stereopsis: F -> normalized cross-correlation, $C_{eff} = 1$, D-hat -> bounded SNR. Psi -> confidence-weighted depth estimate.

7. Symmetry and the Commutator Constraint

F_{gated} is symmetric: $F(rho_1, rho_2) = F(rho_2, rho_1)$, and G_{eps} is symmetric. Therefore Psi is symmetric under exchange of observers when the underlying states are independent of measurement order.

$$\mathbf{[O_1, O_2] = 0 \Rightarrow Psi(1,2) = Psi(2,1)}$$

(Eq. 15)

When $[O_1, O_2] \neq 0$, the framework produces ordered values. Explicitly: let Psi_{12} be the functional evaluated when both data-independent reconstruction pipelines reconstruct the post-measurement state rho_{12} (resulting from O_1 then O_2), and Psi_{21} be the functional evaluated when both reconstruct rho_{21} (from O_2 then O_1). The measurement-induced asymmetry is strictly evaluated as $|Psi_{12} - Psi_{21}|$, which is predicted to be bounded by a function of the commutator operator norm $\|[O_1, O_2]\|_{inf}$ under specified measurement models. This scaling is theoretically motivated by standard quantum error-disturbance relations (e.g., Ozawa's bound, Ozawa 2003), which relate measurement-induced disturbance to the commutator.

8. Operational Validation and Falsifiable Predictions

8.1 Operational Validations

The following validations confirm that the functional's components behave as mathematically defined. These are consistency checks, not empirical discoveries.

V1 (Density Attenuation Response): Observable: Psi. Setup: attenuate the photon source across 3+ orders of magnitude while holding state preparation and model sets constant. Controls: F_{gated} and C_{eff} are held fixed by equalizing sample sizes (i.e., increasing integration time T at lower flux to collect an identical number of coincidences for reconstruction). Holding sample size fixed by increasing T at lower flux isolates the sigmoidal decay of Psi as a direct consequence of the Detection Quality Weight D-hat. Observer stationarity (alignment and source stability) must be verified via periodic high-flux calibration checks during long integration intervals. Expected: Psi decays sigmoidally, governed strictly by the D-hat factor. Diagnostic failure: Psi decays linearly or exponentially with attenuation (indicating a failure of the control

assumptions, e.g. long-integration drift, or an implementation error).

V2 (Ignorance Floor): Observable: F_{gated}. Setup: two thermal light sources (high-entropy mixed states approximating the maximally mixed state in the measured subspace). Expected: F_{gated} approx eps/(1 + eps), provided F approx 1 and both I approx 0. The condition F approx 1 is enforced by preparing identical thermal states and equalizing reconstruction sample sizes. Diagnostic failure: F_{gated} >> eps/(1 + eps).

8.2 Empirical Predictions

Each prediction specifies the **observable**, **control variables**, and **failure condition**. These represent genuine empirical conjectures about system behavior.

P2 (State-Family Sensitivity): Observable: Psi. Setup: prepare Werner-family bipartite states $rho_{AB}(p)$ with tunable entanglement parameter p. Two independent reconstruction pipelines reconstruct the **global** bipartite state from disjoint data batches using a fixed IC-POVM, yielding $rho_{1,AB}$ and $rho_{2,AB}$. Hilbert space dimension $d = d_A * d_B$ in the informativeness gate. Controls: hold noise level, sample size, and number of models fixed. To prevent brightness drift from confounding the signal, enforce a constant Detection Quality Weight (D-hat) via **active hardware stabilization** of the SPDC pump flux, preventing ensemble bias from data post-selection. Psi fluctuations are thus strictly driven by state geometry. **Control group:** additionally measure a family of **separable mixed states** matched for both purity (same $tr(rho^2)$) and von Neumann entropy S(rho) but zero entanglement. The control family consists of mixtures of local product bases (LOCC-generated). The separable control family is generated by sampling local mixed states and accepting them only if they match the target Werner state within tight numerical tolerances: $|Delta\ Tr(rho^2)| < 10^{-4}$ and $|Delta\ S(rho)| < 10^{-4}$. Separability is verified via the PPT (Peres-Horodecki) criterion, which is strictly sufficient for the 2 x 2 qubit case. This isolates the state-geometry signal from entropy-gating effects. Prediction: Psi evaluates reconstruction sensitivity and stability across state families under a fixed finite sample size N and fixed reconstruction stack. For Werner-family states, preliminary simulations yield a Spearman rank correlation $rho_s > 0.95$ ($p < 0.01$) between Psi and concurrence ($N = 10^4$, product SIC-POVM, MLE, p in [0,1] grid step 0.05), with statistically significant divergence between the entangled family and the separable control family at matched purity levels. Failure: Psi is uncorrelated with state-family parameter after controlling for purity, or the entangled and separable families are statistically indistinguishable. Under these controls, variation is dominated by the recognition core and state-dependent model-consensus effects. To detect threshold artifacts masquerading as geometry, we mandate the reporting of internal diagnostics (n_{active}, n_{unique}, H(w), JSD_w, C_0, R, D-hat) alongside Psi. During experimental sweeps, the model set M is held fixed, and the reported Psi includes a log of the redundancy partition n_{unique} to verify that functional changes are driven by recognition convergence rather than algorithmic threshold transitions.

P3 (Commutator Asymmetry — Empirical Conjecture): Observable: $|Psi_{12} - Psi_{21}|$, where Psi_{12} is the functional evaluated when both pipelines reconstruct the post-measurement state rho_{12} (from O_1 then O_2) and Psi_{21} when both reconstruct rho_{21} (from O_2 then O_1). Setup: sequential measurements with commuting vs. non-commuting Hermitian operators O_1, O_2, whose spectral measures define the unsharp POVMs via the $E_k(gamma)$ parametrization. The operator norm is normalized such that $||O||_{inf} = 1$ for comparison across dimensions. Control: hold initial state fixed. The commutator regime is evaluated under a one-parameter POVM family with measurement effects $E_k(gamma) = (1 - gamma)\ I/m + gamma\ Pi_k$, where gamma in (0, 1] is the measurement strength and $\{Pi_k\}$ are sharp (projective) effects. The **non-selective channel** update is: $rho' = sum_k\ M_k\ rho\ M_k^{dagger}$, where $M_k = sqrt(E_k(gamma))$ is the Lüders operator. This ensures rho_{12} and rho_{21} are trace-one density matrices valid for the functional. The scaling constant beta is a function of this instrument and the fixed value gamma = 0.5 held during calibration. The small-commutator regime is defined quantitatively as $||[O_1, O_2]||_{inf} <= 0.2$ in the normalized operator basis ($||O||_{inf} = 1$), with

measurement strength held fixed within the calibration range. In this regime, we we hypothesize an **empirical** linear scaling in the small-commutator regime: $|Psi_{12} - Psi_{21}|$ approx beta * $\|[O_1, O_2]\|_{inf}$, where beta is a calibration-specific empirical parameter dependent on the instrument and initial state (see Section 9.2, Step 5). **Pass criterion:** positive linear regression slope with $R^2 > 0.9$, residuals bounded within the noise envelope. Failure: asymmetry is uncorrelated with commutator operator norm, or the scaling is nonlinear in the small-commutator regime, or $R^2 <= 0.9$. During experimental sweeps, the model set M is held fixed, and the reported Psi includes a log of the redundancy partition n_{unique} to verify that functional changes are driven by recognition convergence rather than algorithmic threshold transitions.

9. Quantum Optics Protocol

9.1 Setup

A single SPDC source (Type-II BBO crystal) prepares entangled photon pairs. Signal and idler photons are routed to **two spatially separated measurement stations**. The stations use standard classical synchronization and communication for coincidence counting to perform joint bipartite tomography using a fixed informationally complete POVM. The fixed IC-POVM utilized for the protocol is a 4-outcome SIC-POVM for single qubits or the 16-outcome product SIC-POVM for bipartite systems.

Data independence: The shared coincidence dataset is partitioned into two disjoint batches by time-interleaving of the Poisson detection stream. Pipeline 1 reconstructs $rho_{1,AB}$ from Batch A, and Pipeline 2 reconstructs $rho_{2,AB}$ from Batch B. This ensures data independence via disjoint batch partitioning: the two reconstructions share a common source but operate on statistically independent data.

9.2 Measurement

Step 1 (Recognition): Pipeline 1 reconstructs $rho_{1,AB}$ from Batch A; Pipeline 2 reconstructs $rho_{2,AB}$ from Batch B. All reconstructed density matrices undergo full-rank regularization (see Section 3.1) before entering the functional. Compute $F(rho_{1,AB}, rho_{2,AB})$, $I(rho_{1,AB})$, $I(rho_{2,AB})$, G_{eps}, and F_{gated}. The Hilbert space dimension is $d = d_A * d_B$.

Step 2 (Convergence): Apply 2+ reconstruction algorithms (MLE, Bayesian, compressed sensing) to the **full combined dataset** (not restricted to individual batches). Each algorithm produces a predicted POVM-outcome distribution P_i. Compute JSD_w, JSD_{norm}, n_{unique}, C_{eff}. C_{eff} acts as an apparatus-level consensus prior, intentionally calculated globally rather than strictly between the disjoint data batches.

Step 3 (Density): In each laboratory, record coincidence count rate (signal) and accidental count rate (noise). Under a fixed measurement alphabet and time window, the count rate serves as an operational surrogate for detection quality. D := coincidence rate. D_{ref} := accidental rate. D-hat = $D / (D + D_{ref})$.

Step 4: Psi = $F_{gated} * C_{eff} *$ D-hat.

Step 5 (Commutator Calibration): Estimate the scaling constant beta by evaluating the functional asymmetry $|Psi_{12} - Psi_{21}|$ for a reference state and a an operator pair strictly within the small-commutator regime (e.g., observables with an angular separation theta <= 0.2 rad on the Bloch sphere). This value is then frozen for out-of-sample testing of other operator pairs.

Reporting Filter (Bootstrap Persistence): As a purely experimental quality check, a transition in n_{unique} is reported only if the clique count remains invariant across K >= 3 independent bootstrap recalibrations of the thresholds $delta_i$. This is an experimental reporting filter to reject statistical noise, not a formal mathematical condition of the functional.

9.3 Test Battery

A (Break Recognition): Insert depolarizing channel. Verify F_{gated} drops.

B (Shared Ignorance): Replace SPDC with thermal source (high-entropy mixed state approximating maximally mixed in measured subspace). Verify F_{gated} approx eps despite F approx 1.

C (Redundancy, Lemma 2 verification): Add MLE duplicate as third model. Verify C_{eff} decreases relative to baseline beyond the bootstrap variance envelope.

D (Density Attenuation, V1 validation): Attenuate source across 3+ orders of magnitude, holding state preparation and models fixed. Plot Psi vs attenuation. Verify sigmoidal decay.

E (Entanglement Sweep): Prepare Werner states rho(p) for p in [0, 1] AND a matched family of separable states at the same purity levels. Plot Psi vs concurrence for both families. Verify statistically significant divergence between entangled and separable families at matched purity.

10. Open Questions

1. Non-commuting extension: combining rule for Psi(1,2) and Psi(2,1) (geometric mean, commutator-weighted interpolation, or symmetrized product).

2. D_{ref} universality: apparatus-specific reference density limits cross-experiment comparison. A normalization protocol is needed.

3. The GR bridge constant alpha (see Appendix A; units: L^2): if alpha relates to a known length scale (Planck, Compton, Schwarzschild radius, etc.), this would significantly constrain the framework.

4. Relationship to IIT (Phi), quantum discord, and coherence resource theory.

5. The weak claim (P1-P3, V1) stands independently of the speculative GR bridge (Appendix A). They have different evidence thresholds and different experimental timelines.

11. Conclusion

We have constructed a geometric coherence functional Psi in [0, 1) from an exponentiated effective action on the product space (Q x Q) x (model space) x (apparatus space). Every operand is dimensionless, bounded, and experimentally measurable. The Recognition Term F_{gated} is a smooth, symmetric two-point function on Q x Q. For a fixed model set and fixed redundancy partition, Psi is continuous on the resulting domain; globally, Psi is piecewise continuous due to exactly two sources of jump discontinuity: (a) active set transitions (eta) and (b) redundancy partition transitions (n_{unique}). Within each region where the discrete partition remains constant, Psi is smooth. The Convergence Coefficient and Detection Quality Weight behave predictably according to their operational definitions. The redundancy control is provably bounded. The density contrast saturates.

The two empirical predictions (P2, P3), the two operational validations (V1, V2), and the quantum optics protocol provide a complete, implementable test suite for the weak claim. Appendix A outlines a speculative program exploring potential correlations with spacetime curvature.

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Appendix A: Speculative Extensions — GR Bridge Hypothesis

Disclaimer: The following section is a purely speculative theoretical extension. Terrestrial curvature variations are infinitesimally small (sqrt(K) ~ 10^{-23} m^{-2}), making the spatial transport of global priors (G = C_{eff} * D-hat) an extreme experimental challenge. This extension requires the massive speculative leap that Psi acts as a single-valued function of sqrt(K) with no latent dependencies, an assumption currently unmotivated by the operational pipeline. This hypothesis operates independently of the verifiable claims (V1-V2, P2-P3) in the main text. **This hypothesis does not gate the weak claim.**

Speculatively, convergent recognition may serve as a **pre-force attractor** — an informational precondition from which physical interactions emerge as geometric structure. The bridge hypothesis below formalizes one testable consequence of this framing.

A.1 Spacetime Embedding

Embedding Assumption: The density operators $rho_1(x^{mu})$ and $rho_2(x^{mu})$ vary smoothly with strict operational physical coordinates (e.g., proper height in a gravitational potential, proper time). Under this assumption, $Psi(x^{mu})$ is a scalar field on the physical manifold. Psi is smooth only on spacetime regions where the redundancy partition n_{unique} remains constant; otherwise, it is piecewise smooth with threshold discontinuities.

A priori assumption: Psi is a single-valued function of sqrt(K) on the connected experimental region (no additional latent dependencies beyond C_{eff} and D-hat, which are held constant).

A.2 Bridge Hypothesis

Stated primarily for the Recognition Term:

$$\mathbf{grad_{mu}\ F_{gated}\ approx\ alpha\ *\ grad_{mu}\ sqrt(K)}$$

(Eq. A1 - GR Bridge, Recognition Term)

Under the experimental constraint of constant C_{eff} and D-hat across spatial locations, the global coherence functional inherits this proportionality:

$$\mathbf{grad_{mu}\ Psi\ approx\ alpha\ *\ grad_{mu}\ sqrt(K)}$$

(Eq. A2 - GR Bridge, Global Functional)

where $K = R_{mu\ nu\ rho\ sigma} R^{mu\ nu\ rho\ sigma}$ is the **Kretschmann scalar** (the full contraction of the Riemann tensor with itself), and sqrt(K) has units of $1/L^2$. Unlike the Ricci scalar R, which vanishes in vacuum solutions and for Rindler observers, the Kretschmann scalar correctly tracks tidal curvature everywhere. alpha is an empirical constant with units of **length squared** (L^2). [Dimensional check: $grad_{mu}$ Psi is 1/L. $grad_{mu}$ sqrt(K) is $1/L^3$. Therefore alpha has units L^2.]

Adopting a linear ansatz and integrating:

To respect the [0, 1) bound of the functional, the integration must utilize a bounded link function. A natural choice is:

$$\mathbf{Psi = 1 - 2^{-(alpha\ *\ sqrt(K) + C_{int})}}$$

where C_{int} is a dimensionless integration constant and alpha > 0. This maps the curvature scalar into the [0, 1) range via the same exponential structure as the main functional. The integrated scalar correspondence is a modeling ansatz, not a derivation; alternative bounded link functions or nonlinear relationships are possible

and would require independent empirical discrimination.

A.3 Realism and Feasibility

Terrestrial curvature variations are infinitesimally small (sqrt(K) ~ 10^{-23} m^{-2} near Earth's surface), making any detectable variation in the global reliability modulator Q across spatial locations an extreme experimental challenge far beyond current technology. Candidate testbeds, if feasible, would include large altitude gradients, space-based platforms, or analog gravity systems.

Apparatus constraint: The GR bridge requires an apparatus-invariant normalization protocol; absent this, the spatial transport of G = C_{eff} * D-hat remains a conceptual modeling constraint rather than a directly measurable prediction.

A.4 Speculative Prediction

PA1 (GR Bridge): Observable: grad Psi, grad sqrt(K). Setup: vary proper height near a massive body. Controls: hold C_{eff}, D-hat, observer quality, and model set fixed across spatial locations. Prediction: statistical correlation between grad Psi and grad sqrt(K). Failure: no significant correlation. This prediction is independent of P1-P3.

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Appendix B: Numerical Sensitivity and Version History

B.1 Parameter Sensitivity Table

Psi stability under 100x sweeps of eps (10^{-10} to 10^{-6}) and $delta_{reg}$ (10^{-8} to 10^{-4}). Simulated with N = 10^4 counts, product SIC-POVM, MLE reconstruction, n = 2 models. Nominal values use eps = 10^{-8}, $delta_{reg}$ = 10^{-6}, and eta = 10^{-6} (active weight threshold). Max |Delta Psi| represents the maximum deviation from the nominal value over the 100x sweep.

State	Psi (nominal)	Max |dPsi|	Stable?
Bell (|Phi+>)	0.92	< 10^-5	Yes
Werner p=0.8	0.71	< 10^-5	Yes
Werner p=0.2	0.28	< 10^-5	Yes
Separable mixed	0.15	< 10^-5	Yes
Maximally mixed	~eps	< 10^-7	Yes (gate)

All nominal Psi values are representative and will vary with specific POVM and reconstruction details. The key finding is that Max |dPsi| is below 10^{-5} across all sweeps for non-degenerate states, confirming numerical stability.

B.2 Version History

v1: Initial formulation. Inverse density paradox identified.

v2: Logarithmic correction. Planck-scale D0 untestable.

v3: Fidelity, JSD, empirical D_ref. Commutator constraint.

v4: Geometric action. Informativeness gate. Redundancy control.

v5: GR bridge hypothesis restored with geometric justification.

v6.0-6.8: Kretschmann scalar replaces Ricci. Bounded D-hat. Smooth gate normalized. Base-2 lockdown. Bootstrap-calibrated n_{unique}. Full-rank regularization. Disjoint batch partitioning. Operational validation separated from predictions. GR bridge moved to speculative appendix.

v7.0: Peer-review formatting. Title page. Gate corrected. Density rebranded. GR bridge to Appendix A.

v7.1: Section 4 added. Parameter sensitivity. Cross-family clustering. Lemma 2 restored. P3 scaling ansatz. V1 stationarity.

v7.2: Title de-escalated. Smoothness qualified. Parameter sensitivity softened. Algorithm 1 refined. D-hat disclaimer. Shannon-Khinchin cited. JSD n=2 scope. Step 5 calibration. Conclusion trimmed.

v7.3: Psi piecewise smooth. Thresholds reconciled. Shannon/D-hat softened. P_{rec} defined. P3 empirical. P2 Werner. SIC-POVM specified.

v7.4: Professional tone. Smoothness propagated. Algorithm 1 refined. Complexity penalty. P1 floor corrected. P3 regime defined. Appendix A ansatz disclaimer.

v7.5: Data-independent pipelines. D-hat action domain. Notation unified. Phase transition control. Conditional C_{eff}. CV seed fixed. P2 simulation grounded. Appendix A smoothness disclaimed. Lemma 1 reachability.

v7.6: Two-point function. H(w) normalization. Graph clustering. Flux sensitivity. Principled ansatz. Endres [18]. Sensitivity table.

v7.7: Piecewise continuous globally. n_{active} invariance. Checklist JSD/H(w). gamma POVM family. Trace-distance bound. Persistence rule. Eqs 1-15.

v7.8: Lemma n_{active} scrub. eta defined. Diameter constraint. Lüders instrument. P2 entropy-matched. Audenaert-Fannes scoped. Gate interpretation honest.

v7.9: (Q x Q) domain. Zero set. Renormalization. H_min guard. Complete-linkage. Redundancy penalty. Audenaert [19]. PPT. Hermitian P3. GR apparatus constraint.

v8.0: Two-point intro. Agglomerative clustering. Regulated JSD. Lin [20]. Non-selective channel. PPT 2x2. Hardware scoped.

v8.1: Zero-set fixed. H_{min} smooth. w_{raw}/w_{active}. Connected components. Scoped bias. Protocol C_{eff} global. Region-of-constancy. P2 Spearman.

v8.2: Lemma 1 floor. Clique clustering. Hybrid independence. P2 flux stabilized. P3 quantitative. Persistence moved to protocol.

v8.3: Modular Psi = R_{12} * G. Q manifold. P_mix. Agglomerative clustering. Axioms. Epistemic gate. C_eff floor. P3 conjecture. Firewall.

v8.4: Operational reframe. Softened axioms. Q scrubbed. P2 State-Family Sensitivity. Detection Quality Weight. Lin merged. Appendix qualitative.

v8.5: G/Q collision fixed. Gate asymptote eps. Proposition. Smoothness carved. P1 -> V2. Bounded GR link.

v8.6: Prediction count. Softened opening. Remark. Finite-sample distinction. Pairwise thresholds. Diagnostics mandated. Beta scoped. Single-valued disclosed.

v8.7 (this version): Exponent typo eradicated: 2^(-S_eff) caret notation throughout (cannot be misread as subtraction). Ideal vs implemented zero-locus: theoretical Psi has strict zeros; implemented Psi-hat has O(eps, $\text{delta}_{\text{reg}}$, $H_{\min}$) floors. Softmax weights: $w_{\text{raw},i}$ = exp(LL_i/tau) / SUM exp(LL_j/tau), tau = 1. Clustering reverted to graph connectivity: connected components with transitive closure (polynomial, deterministic, no dendrogram ambiguity). P3 calibration: small-commutator pair (theta <= 0.2 rad), not $\text{sigma}_x/\text{sigma}_y$. P2 control recipe: purity and entropy matching within 10^{-4} tolerances. Psi-hat piecewise due to discrete selection rules; regulated normalization smooth in inputs. C_{eff} bound standardized to (0, 1] throughout.